

Enterprise Cisco / Arista Switch Deployment using Zero Touch Provisioning (ZTP)

Version: 1.0

Environment: Enterprise Data Center

Deployment Method: Zero Touch Provisioning (ZTP)

Automation: Ansible / Ansible Automation Platform (AAP)

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Overview

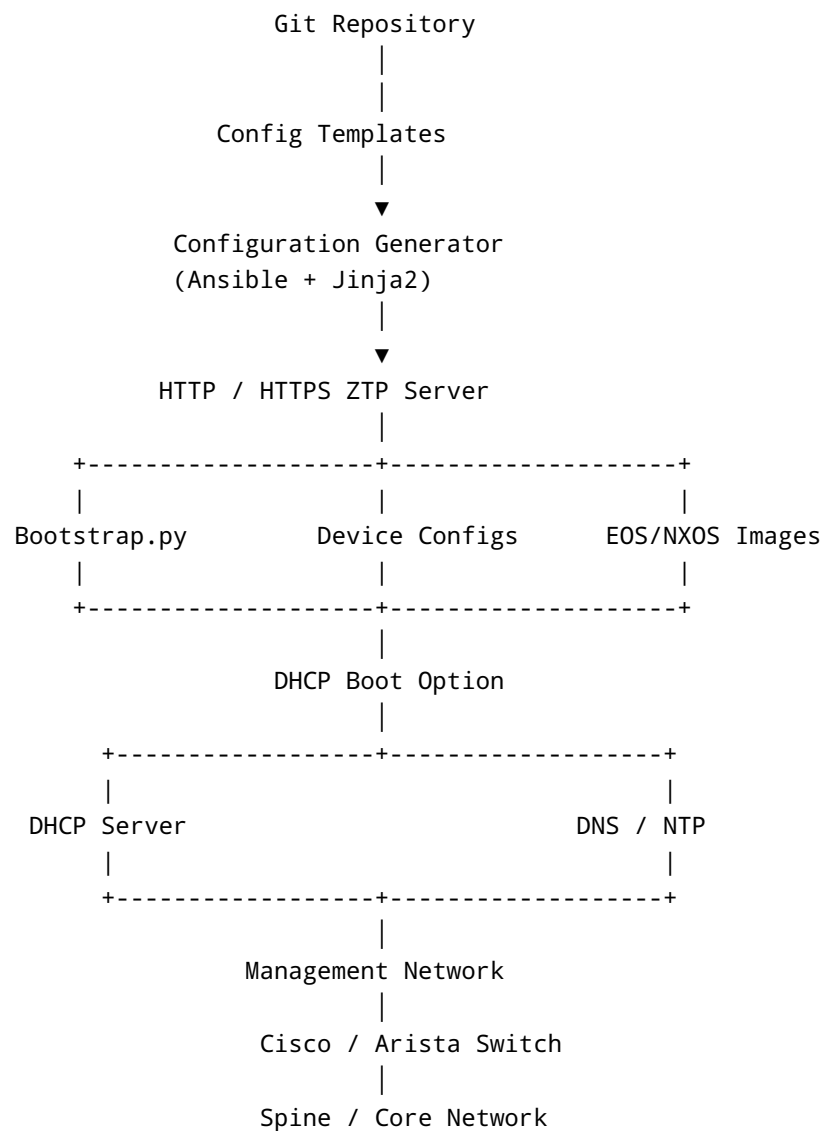
This repository describes the complete enterprise workflow for deploying Cisco Nexus and Arista EOS switches using Zero Touch Provisioning (ZTP).

The deployment is fully automated and requires no manual console configuration after the switch is racked and powered on.

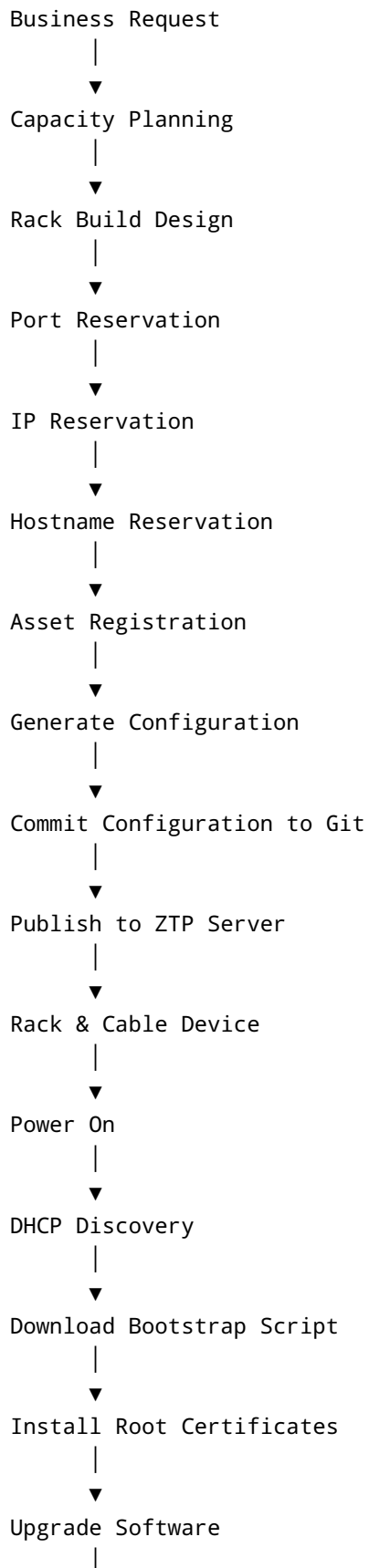
The provisioning process includes:

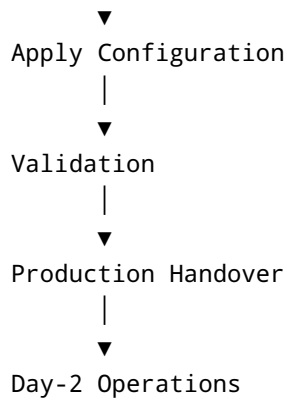
- Rack build
- Port reservation
- Asset registration
- IP allocation
- Configuration generation
- Root CA installation
- Software upgrade
- Device provisioning
- Validation
- Production handover

Enterprise Architecture



End-to-End Deployment Flow





Infrastructure Components

Component	Purpose
Git	Version control
Ansible	Configuration generation
Jinja2	Templates
DHCP	IP assignment
HTTP/HTTPS	Bootstrap server
PKI	Certificates
DNS	Name resolution
NTP	Time synchronization
Syslog	Logging
TACACS+/RADIUS	Authentication
Prometheus	Monitoring
Grafana	Dashboards

Phase 1 – Project Request

The infrastructure request includes:

- Number of switches
- Rack location
- Server count
- Network speed
- VLAN requirements
- Routing requirements

- Redundancy requirements

Deliverables:

- Approved design
- Bill of Materials
- Change Request

Phase 2 – Rack Build

Example Rack Layout

```
U42 Patch Panel
U41 Patch Panel
U40 Spine-01
U39 Spine-02
U38 Blank
U37 Leaf-01
U36 Leaf-02
U35 ESXi-01
U34 ESXi-02
U33 Kubernetes-01
```

Each device receives:

- Rack Number
- RU Position
- Power Feed A
- Power Feed B
- Cable ID

Phase 3 – Asset Registration

Field	Example
Vendor	Arista
Model	7050SX3
Serial Number	JPE123456789
MAC Address	3C:2C:30:AA:BB:CC
Rack	R101
RU	U37

Asset information is stored in the CMDB.

Phase 4 – Port Reservation

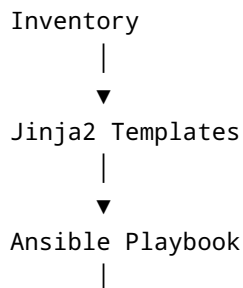
Port	Connected To
Mgmt1	Management Switch
Ethernet1	ESXi01
Ethernet2	ESXi02
Ethernet49	Spine01
Ethernet50	Spine02
Ethernet51	MLAG Peer
Ethernet52	MLAG Peer

Phase 5 – IP Reservation

Purpose	Address
Management	10.10.1.21
Loopback	172.16.1.21
VTEP	192.168.100.21
Gateway	10.10.1.1
DNS	10.10.1.10
NTP	10.10.1.20

Phase 6 – Configuration Generation

Configurations are generated automatically.



▼
Device Configuration

Example output:

- Hostname
- Interfaces
- VLANs
- MLAG
- BGP
- VXLAN
- ACLs
- SNMP
- Logging

Phase 7 – Git Repository

```
network-switch/  
  
README.md  
  
inventory/  
  
templates/  
  
playbooks/  
  
ztp/  
  
bootstrap/  
  
images/  
  
configs/  
  
certificates/  
  
docs/
```

Every configuration change is committed through Git.

Phase 8 – ZTP Server

```
ztp/  
  
bootstrap.py  
  
configs/  
  
images/  
  
certificates/  
  
logs/
```

Services:

- DHCP
- HTTP
- HTTPS

Phase 9 – PKI & Certificate Installation

Why?

Before the switch communicates with enterprise services, it must trust the corporate PKI.

Certificates installed:

```
RootCA.pem  
  
IntermediateCA.pem
```

Trusted services:

- Git
- ZTP Server
- TACACS+
- Grafana
- Prometheus
- HTTPS APIs

Certificate workflow:

```
Download Root CA
```


↓

Import Trust Store

↓

Verify Certificate

↓

Continue Bootstrap

Phase 10 – Switch Boot

Power On

↓

Boot Loader

↓

Load Operating System

↓

No Startup Config

↓

Start ZTP

Phase 11 – Bootstrap Workflow

DHCP Discover

↓

Receive Management IP

↓

Receive Bootstrap URL

↓

Download bootstrap.py

↓

Read Serial Number

↓

Install Root CA

↓

Verify HTTPS

↓

Download Configuration

↓

Upgrade Image

↓

Apply Configuration

↓

Save Configuration

Phase 12 – Software Upgrade

Current Version

↓

Compare Approved Version

↓

Download Image

↓

Install

↓

Reload

↓

Continue Bootstrap

Phase 13 – Configuration Applied

Bootstrap configures:

- Hostname
- Users
- SSH
- Management Interface
- DNS
- NTP
- Syslog
- TACACS+
- SNMP
- Telemetry
- Interfaces
- VLANs
- Port Channels
- MLAG/vPC
- BGP
- EVPN
- VXLAN
- ACLs

Phase 14 – Monitoring

Switch is registered with:

- Prometheus
 - Grafana
 - Syslog
 - Configuration Backup
 - Automation Inventory
-

Phase 15 – Validation

Validation checklist:

- Management reachable
 - SSH accessible
 - Correct hostname
 - Correct software version
 - Root CA installed
 - TACACS login successful
 - DNS working
 - NTP synchronized
 - LLDP neighbors detected
 - Uplinks operational
 - BGP established
 - Configuration saved
-

Phase 16 – Production Handover

Operations team receives:

- Device Information
 - Configuration Backup
 - Monitoring Status
 - Validation Report
 - Change Record
 - Asset Details
-

Phase 17 – Day-2 Operations

Supported operations:

- Add VLAN
 - Add Server Ports
 - Firmware Upgrade
 - Certificate Renewal
 - Backup
 - Compliance Check
 - Interface Changes
 - RMA Replacement
 - Monitoring Alerts
-

Rollback Procedure

1. Stop traffic migration.
 2. Restore previous switch.
 3. Restore previous configuration.
 4. Verify application connectivity.
 5. Analyze deployment logs.
 6. Correct issue.
 7. Retry deployment.
-

Repository Structure

```
network-switch-deployment/  
  
README.md  
  
docs/  
  
inventory/  
  
templates/  
  
playbooks/  
  
bootstrap/  
  
configs/  
  
images/  
  
certificates/  
  
scripts/  
  
validation/  
  
backup/  
  
monitoring/
```

Security Best Practices

- Use HTTPS only.
- Disable Telnet.
- Use TACACS+/RADIUS for AAA.

- Install Root and Intermediate CA before downloading configuration.
 - Store all configurations in Git.
 - Rotate certificates before expiry.
 - Sign software images where supported.
 - Encrypt sensitive Ansible variables with Ansible Vault.
-

Future Enhancements

- NetBox integration
- ServiceNow integration
- Automated IPAM allocation
- GitOps workflow
- CI/CD validation
- Automated compliance scanning
- Automatic certificate renewal
- Image lifecycle management
- Drift detection
- Self-service switch provisioning